Chapter 4 Answers

Problem 1

(a): We suppose that R_1 and R_2 are of the same degree (n) and are both formally representable over some alphabet K. Hence, by Proposition 1, there is an EFS (call it E) that is common to R_1 and R_2 under which the both of them are formally representable under the predicates P_1 and P_2, respectively. Now, add to E the axioms (with a new predicate P):

$$P_1x_1, ..., x_n \rightarrow Px_1, ..., x_n$$

$$P_2x_1, ..., x_n \rightarrow Px_1, ..., x_n$$

which represent the relation $R_1 \bigcup R_2$. To represent the relation $R_1 \bigcap R_2$, add the axiom (with a new predicate P_0):

$$P_1x_1, ..., x_n \rightarrow P_2x_1, ..., x_n \rightarrow P_0x_1, ..., x_n$$

(b): We suppose that $R(x_1, ..., x_n, y)$ is a relation of degree $n+1$ that is formally representable over K (by some predicate P). To represent the relation $\exists yR(x_1, ..., x_n, y)$ add the axiom (with a new predicate P_0):

$$Px_1, ..., x_n, y \rightarrow P_0x_1, ..., x_n$$

(c): Suppose that $R(x_1, ..., x_k)$ is a relation of degree k that is formally representable over K (by some predicate P). To represent the relation $\lambda x_1, ..., x_nR(\alpha_1, ..., \alpha_k)$, add the axiom (with a new predicate P_0):

$$P\alpha_1, ..., \alpha_k \rightarrow P_0x_1, ..., x_n$$

To see this, recall that each α_i is either x_i or some constant. Hence, if $P\alpha_1, ..., \alpha_k$ is provable (that is, $R(\alpha_1, ..., \alpha_k)$ holds), it is provable after substituting some constant term c in for every variable x_i corresponding to an α_i, while leaving every constant α_i as is. Now $\lambda a_1, ..., a_nR(\alpha_1, ..., \alpha_k)$ holds for an n-tuple $a_1, ..., a_n$ iff $R(b_1, ..., b_k)$ holds where, if α_i is a constant, then b_i is α_i, whereas

if α_i is x_i, then b_i is a_i. So in the added axiom, the $P_0 x_1, ..., x_n$ part is actually evaluating a given n-tuple $a_1, ..., a_n$, and for every $\alpha_i = x_i$, α_i becomes a_i (in the *instance* of the axiom for the n-tuple $a_1, ..., a_n$, which we are considering). Hence $P\alpha_1, ..., \alpha_k$ means $R(\alpha_1, ..., \alpha_k)$ holds where if α_1 was initially a constant, it stays the same, whereas if it was a variable, it becomes a_i. But this means that $P_0 x_1, ..., x_n$ doesn't actually change the values of any b_i from the corresponding value in the above instance of $Px_1, ..., x_n, y$ (because all of them have been changed to constants at this point that are the same as their corresponding b_i's), meaning $R(b_1, ..., b_k)$ holds as well.

Problem 2

(a): Suppose that both R_1 and R_2 are solvable over some EFS over K. This means that all of $R_1, R_2, \widetilde{R_1}, \widetilde{R_2}$ are representable over K. Hence $R_1 \bigcup R_2$ is representable over K by Problem 1(a). $\widetilde{R_1 \bigcup R_2}$ is the same as $\widetilde{R_1} \bigcap \widetilde{R_2}$, and because both $\widetilde{R_1}$ and $\widetilde{R_2}$ are representable over K, $\widetilde{R_1} \bigcap \widetilde{R_2}$ (that is, $\widetilde{R_1 \bigcup R_2}$) is also representable over K (by Problem 1(a)). Now, $R_1 \bigcap R_2$ is formally representable over K by Problem 1(a). $\widetilde{R_1 \bigcap R_2}$ is the same as $\widetilde{R_1} \bigcup \widetilde{R_2}$. Because both $\widetilde{R_1}$ and $\widetilde{R_2}$ are representable over K, so is $\widetilde{R_1} \bigcup \widetilde{R_2}$ (by Problem 1(a)).

(b): Suppose that R (of degree k) is solvable over K. Hence both R and $\bar{R}$ are formally representable over K, which means that $\lambda x_1, ..., x_n R(\alpha_1, ..., \alpha_k)$ and $\lambda x_1, ..., x_n \bar{R}(\alpha_1, ..., \alpha_k)$ are both formally representable over K (by Problem 1(c)). Now, $\lambda x_1, ..., x_n R(\alpha_1, ..., \alpha_k)$ is the complement of $\lambda x_1, ..., x_n \bar{R}(\alpha_1, ..., \alpha_k)$. To see this, note that, for any n-tuple $a_1, ..., a_n$, $\lambda a_1, ..., a_n R(\alpha_1, ..., \alpha_k)$ holds iff $R(b_1, ..., b_k)$ holds, and $\lambda a_1, ..., a_n \bar{R}(\alpha_1, ..., \alpha_k)$ holds iff $\bar{R}(b_1, ..., b_k)$ holds iff $R(b_1, ..., b_k)$ does not hold iff $\lambda a_1, ..., a_n R(\alpha_1, ..., \alpha_k)$ does not hold, which means that for all $x_1, ..., x_n$, $\lambda x_1, ..., x_n R(\alpha_1, ..., \alpha_k)$ holds iff $\lambda x_1, ..., x_n \bar{R}(\alpha_1, ..., \alpha_k)$ does not hold, thus showing that they are the complement of one another.

Problem 3

(a): Suppose that $R(x_1, ..., x_n, y)$ is recursively enumerable (by some predicate P). On the one hand, consider $\forall_F R$. This is the relation

$$(\forall z \leq y)\, R(x_1, ..., x_n, z)$$

This relation can be represented by adding the axioms (for some new predicate Q):

$$Px_1, ..., x_n, 1 \to Qx_1, ..., x_n, 1$$

$$Qx_1, ..., x_n, z \to Sz', z \to Px_1, ..., x_n, z' \to Qx_1, ..., x_n, z'$$

where Sz', z means "z' is the successor of z" (which was shown to be recursively enumerable in *The Beginner's Guide*). As for $\exists_F R$, this can be represented by the axiom (for some new predicate Q_1):

$$z \leq y \to Px_1, ..., x_n, z \to Qx_1, ..., x_n, y$$

(b): Suppose $R(x_1, ..., x_n, y)$ is recursive. Hence both $R(x_1, ..., x_n, y)$ and $\bar{R}(x_1, ..., x_n, y)$ are recursively enumerable, and hence, by Problem 3(a), both $\forall_F R$ and $\forall_F \bar{R}$ ($\exists_F R$ and $\exists_F \bar{R}$) are recursively enumerable. But $\forall_F R$ and $\exists_F \bar{R}$ ($\exists_F R$ and $\forall_F \bar{R}$) are complements, meaning $\forall_F R$ ($\exists_F R$) is recursive.

Problem 4

Suppose that we are given some Σ_1 relation $R(x_1, ..., x_n, y)$. Now, it is obvious that:
(1): $\exists y \exists z R(x_1, ..., x_n, y, z)$
(2): $\exists x (\exists y \leq x) (\exists z \leq x) R(x_1, ..., x_n, y, z)$
are equivalent for any relation $R(x_1, ..., x_n, y, z)$. Well, because $R(x_1, ..., x_n, y)$ is Σ_1, it is equivalent to some relation $\exists z S(x_1, ..., x_n, y, z)$ where $S(x_1, ..., x_n, y, z)$ is Σ_0. Hence $\exists y R(x_1, ..., x_n, y)$ is equivalent to $\exists y \exists z S(x_1, ..., x_n, y, z)$ (See this in the following way: $\exists y R(x_1, ..., x_n, y)$ holds iff $R(x_1, ..., x_n, y)$ holds for some y iff $\exists z S(x_1, ..., x_n, y, z)$ holds (for the same y) iff $\exists y \exists z S(x_1, ..., x_n, y, z)$ holds), a formula that is in turn equivalent (by the first reasoning in this solution) to one of the form

$$\exists x (\exists y \leq x) (\exists z \leq x) S(x_1, ..., x_n, y, z)$$

that is, one of the form

$$\exists z R(x_1, ..., x_n, z)$$

where $R(x_1, ..., x_n, z)$ is Σ_0 (taking $R(x_1, ..., x_n, z)$ to be $(\exists y \leq x) (\exists z \leq x) S(x_1, ..., x_n, y, z)$) thus meaning $\exists y R(x_1, ..., x_n, y)$ is also Σ_1.

Problem 5

We suppose that A is a Σ_1 set and $f(x)$ is a Σ_1-function, and that A' is the set of all x such that if $f(x) \in A$. Now, A being a Σ_1 set means it is

expressed by some formula of the form $\exists z R(w,z)$ where $R(w,z)$ is a Σ_0 relation (it has to be a binary relation as the relation of being in a set requires only one argument), and $f(x)$ being a Σ_1-function means $f(x)=y$ is a Σ_1 relation, and hence it is expressed by some formula of the form $\exists r S(x,y,r)$ where $S(x,y,r)$ is a Σ_0 relation. Consider the relation

$$f(x)=y \wedge x \in A$$

which is equivalent to

$$\exists z R(w,z) \wedge \exists r S(x,y,r)$$

which is equivalent to

$$\exists z \exists r \left(R(w,z) \wedge S(x,y,r)\right)$$

which is equivalent to (by Problem 4):

$$\exists q \left(\exists z \leq q\right)\left(\exists r \leq q\right)\left(R(w,z,) \wedge S(x,y,r)\right)$$

which is a Σ_1 function. Hence $f(x)=y \wedge x \in A$ is Σ_1, therefore meaning the set A' is Σ_1.

Problem 7

Recall that, for the function $\delta(x,y)$:
(1): $\delta(x,y)=(x-1)^2+x$ if $x \leq y$
(2): $\delta(x,y)=(x-1)^2+x+y$ if $y<x$
both hold. Hence $\delta(x,y)=z$ is equivalent to:

$$(1)\ \ (\exists w \leq z)\left[\left(x \leq y \supset z=w^2+x\right) \wedge \left(y<x \supset z=w^2+x+y\right)\right]$$

Recall that we have already shown that $x^y=z, x+y=z, x \leq y, y<x$ are all Σ_0 relations. Hence (1) is Σ_0, thus proving that $\delta(x,y)$ is Σ_0. This obviously means that $\sim\delta(x,y)$ is also Σ_0, thus meaning both $\delta(x,y)$ and $\sim\delta(x,y)$ are recursively enumerable, and hence $\delta(x,y)$ is a recursive function.

Problem 8

(a): This is completely obvious, as

$$\delta(Kz,Lz)=\delta(x,y)=z$$

(b): $K(z)=x$ is equivalent to $(\exists y \leq z)\left(\delta(x,y)=z\right)$ which is Σ_0. $L(z)=y$ is equivalent to $(\exists x \leq z)\left(\delta(x,y)=z\right)$ which is Σ_0.

Problem 9

(a): We consider some recursively enumerable set A and $n \geq 2$. Consider the relation: $\delta_n(x_1, ..., x_n) \in A$. Now, it has previously been shown that $\delta_n(x_1, ..., x_n)$ is a recursive function, and hence it is Σ_1 of the form:

$$\exists z R(x_1, ..., x_n, x, z)$$

for some Σ_0 relation $R(x_1, ..., x_n, x, z)$. By the assumption that A was recursively enumerable, it is Σ_1 of the form:

$$\exists w S(y, w)$$

for some Σ_0 relation $S(y, w)$. Now, $\delta_n(x_1, ..., x_n) \in A$ is equivalent to

$$\delta_n(x_1, ..., x_n) = y \wedge y \in A$$

which is equivalent to

$$\exists z R(x_1, ..., x_n, x, z) \wedge \exists w S(y, w)$$

which is equivalent to

$$\exists z \exists w (R(x_1, ..., x_n, x, z) \wedge S(y, w))$$

which is Σ_1, thus proving $\delta_n(x_1, ..., x_n) \in A$ is recursively enumerable.

(b): Consider some recursively enumerable relation $R(x_1, ..., x_n)$. Hence it is Σ_1 of the form:

$$\exists z S(x_1, ..., x_n, z)$$

for some Σ_0 relation S. Now, a number y is in the set of all numbers $\delta_n(x_1, ..., x_n)$ such that $R(x_1, ..., x_n)$ iff

$$\delta_n(x_1, ..., x_n) = y \wedge R(x_1, ..., x_n)$$

for some n-tuple $x_1, ..., x_n$ which is equivalent to:

$$\exists x_1, ..., \exists x_n (\exists w P(x_1, ..., x_n, y, w) \wedge \exists z S(x_1, ..., x_n, z))$$

(taking $\exists w P(x_1, ..., x_n, y, w)$ to be the Σ_1 form of $\delta_n(x_1, ..., x_n)$) which is equivalent to:

$$\exists x_1, .., \exists x_n \exists w \exists z (P(x_1, ..., x_n, y, w) \wedge R(x_1, ..., x_n))$$

which is Σ_1.

Problem 10

(a): If $\delta_n(x_1, .., x_n) = x$, then $K_1^n(x) = x_1, K_2^n(x) = x_2, ..., K_n^n(x) = x_n$. Thus

$$\delta_n(K_1^n(x), K_2^n(x), ..., K_n^n(x)) = \delta_n(x_1, x_2, ..., x_n) = x$$

(b): Suppose that $\delta(x_1, x_2) = x$. Hence $K_1^2(x) = Kx = x_1$ and $K_2^2(x) = Lx = x_2$.

(c): Suppose $x = \delta_{n+1}(x_1, ..., x_{n+1})$. By definition, this means that

$$x = \delta(\delta_n(x_1, ..., x_n), x_{n+1})$$

and thus $Kx = \delta_n(x_1, ..., x_n)$ and $Lx = x_{n+1}$.

(d): Suppose that $\delta_{n+1}(x_1, ..., x_{n+1}) = x$. Then, by definition, $\delta(\delta_n(x_1, ..., x_n), x_{n+1}) = x$, thus meaning $K_i^n(Kx) = K_i^n(\delta_n(x_1, ..., x_n)) = x_i = K_i^{n+1}$, if $i \leq n$, while $Lx = x_{n+1} = K_i^{n+1} = x_{n+1}$ if $i = n + 1$.

(e): Consider some recursively enumerable relation $R(x_1, ..., x_n, y_1, ..., y_m)$. Hence it is Σ_1 of the form $\exists z S(x_1, ..., x_n, y_1, ..., y_m, z)$ for some Σ_0 relation S. We take $M(x, y)$ to be the relation:

$$\exists z S(K_1^n(x), ..., K_n^n(x), K_1^m(y), ..., K_m^m(y), z)$$

which is Σ_1.

Problem 11

(a): Add a new predicate N:

$$N1$$

$$N2$$

$$Nx \to Ny \to Nxy$$

(b): Add a new predicate A:

$$A'$$

$$Ax \to Ax'$$

(c): Add a new predicate V:

$$Ax \to Vvxv$$

(d): Add a new predicate D:

$$Vvxv \to Ay \to Dvxv, vxyv$$

$$Dx, y \to Dy, x$$

(e): Add a new predicate P:

$$Ax \to Ppx$$

$$Px \to Ppx$$

(f): Add a new predicate T:

$$Nx \to Tx$$

$$Vx \to Tx$$

$$Tx \to Ty \to Txy$$

(g): Add a new predicate F_0:

$$Tt \to Ax \to F_0pxt$$

$$Tt \to F_0x \to F_0pxcomt$$

(h): Add a new predicate F:

$$F_0x \to Fx$$

$$Fx \to F_0y \to Fximpy$$

(i): Add a new predicate S_0:

$$Ax \to Ny \to S_0pxy$$

$$S_0x \to Ny \to S_0pxcomy$$

(j): Add a new predicate S (for TS sentence, a problem he apparently forgot to add):

$$S_0x \to Sx$$

$$Sx \to S_0y \to Sximpy$$

(k): Add a new predicate S_2:

$$Nx \to Vy \to Nz \to S_2x, y, z, x$$

$$Vx \to Dx, y \to Nz \to S_2x, y, z, x$$

$$Vx \to Nz \to S_2x, x, z, z$$

$$Vy \to Nz \to S_2com, y, z, com$$

$$Vy \to Nz \to S_2imp, y, z, imp$$

$$Vy \to Nz \to Pp \to S_2p, y, z, p$$

$$S_2x, y, z, w \to S_2x_1, y, z, w_1 \to S_2xx_1, y, z, ww_1$$

(l): Add a new predicate I_0:

$$Fx \to S_2x, y, z, w \to I_0x, w$$

$$I_0x, y \to I_0y, z \to I_0x, z$$

(m): Add a new predicate I:

$$I_0x, y \to Sy \to Ix, y$$

(n): Add a new predicate B:

$$Fx \to Bx$$

$$Bx \to Fy \to Bx \wedge y$$

(o): Add a new predicate C:

$$Fx \to Cx, x$$

$$Cx, y \to Bz \to Cx, y \wedge z$$

$$Cx, y \to Bz \to Cx, z \wedge y$$

(p): Add a new predicate T:

$$Cy, x \to Iy, z \to Tx \vdash z$$

$$Tx \vdash y \to Tx \vdash yimpz \to S_0y \to Tx \vdash z$$

Note: S_0x needs to be added as the definition of provability says X must be an atomic sentence if Y is inferred from X and $X \to Y$.

Problem 12

First, the relation "the Gödel number of x is y" is recursively enumerable under the axioms:

$$G1, 12$$

$$G2, 122$$

$$Gx, y \rightarrow Gq, r \rightarrow Gxq, yr$$

Hence, to represent the relation "the norm of x is y", we add the axiom:

$$Gx, y \rightarrow Nx, xy$$

Note: Smullyan seems to have made a mistake in his answer. When he was giving axioms for the relation x has the Gödel number y, he was treating it like a unary relation (saying Gx and Gy), even though he defined it as a binary relation.

Problem 13

Suppose that A is recursively enumerable. Hence it is Σ_1 of the form $\exists z R(x, z)$ for some Σ_0 relation $R(x, z)$. Now, a number $x \in A^{\#}$ iff the norm of $x \in A$. Suppose $\exists p S(q, r, p)$ is the relation that is Σ_1 for r being the norm of q (as shown to be Σ_1 in Problem 12). Hence $q \in A^{\#}$ iff:

$$\exists p \exists r S(q, r, p) \wedge \exists z R(r, z)$$

which is Σ_1.

Alternatively: Suppose that A is represented by the predicate P. Then, for some new predicate M:

$$Nx, y \rightarrow Py \rightarrow Mx$$

M represents the set $A^{\#}$.

Problem 14

Now suppose that A is recursively enumerable. Hence, by Problem 13, $A^{\#}$ is also recursively enumerable. Suppose that the predicate H represents $A^{\#}$, and that it has a Gödel number of n. Thus Hn is true iff $n \in A^{\#}$ iff $nn_0 \in A$ (for n_0 the Gödel number of n), thus proving that Hn is a Gödel sentence for A, as nn_0 is the Gödel number of Hn.